



forestry, fisheries & the environment

Department:
Forestry, Fisheries and the Environment
REPUBLIC OF SOUTH AFRICA

DEPARTMENT OF FORESTRY, FISHERIES and the ENVIRONMENT BRANCH: FISHERIES MANAGEMENT SMALL PELAGIC SCIENTIFIC WORKING GROUP

RESULTS OF THE 2025 PELAGIC RECRUIT SURVEY

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Summary

Recruit estimates of anchovy, sardine and round herring for the area between the Orange River and Cape Infanta are presented. These indicate record low recruitment for anchovy but improved sardine recruitment and above average recruitment for round herring in the area between the Orange River and Cape Infanta.

The 2025 pelagic recruit survey was conducted aboard the Fisheries Research Ship *Africana*. The survey was started and completed (i.e., 09-29 July) within the time window of previous surveys. The survey covered most of the inshore areas along the South African coastline from the mouth of the Orange River on the west coast to Cape Infanta on the south coast. Survey work included acoustic surveying, trawling for species identification, and biological sampling of fish. No environmental sampling was conducted.

The survey started slightly later than planned due to financial constraints, certification requirements and last minute malfunction of one of the converters. A total of 25 days were set aside to complete the whole survey; however, 20 days were used as the ship's engines performed well in the cooler inshore waters. Relatively good weather conditions were encountered during most of the survey and only 7 hours survey time was lost due to bad weather conditions, and about 9 hours due to minor technical issues with the trawl winch.

In total, 58 survey transects were sampled from the Orange River to Cape Infanta and 78 target identification trawls were conducted. A total of 3093 nautical miles (nmi) were steamed to complete these transects. The cruise track and location of pelagic trawls are presented in Figure 1. The coverage is slightly less than in recent years (71 transects west of Cape Infanta in 2024 compared to 58 in 2025) given slightly reduced survey effort especially in strata A and B due to limited survey time. The survey was mainly limited to the inshore areas and most transects only extended about 20 to 30 nmi offshore reflecting the near-shore distribution of recruit fish. The coverage is considered adequate and the amount of trawling for target identification and length frequency sampling in high-density areas is good and deemed satisfactory for anchovy, sardine and redeye.

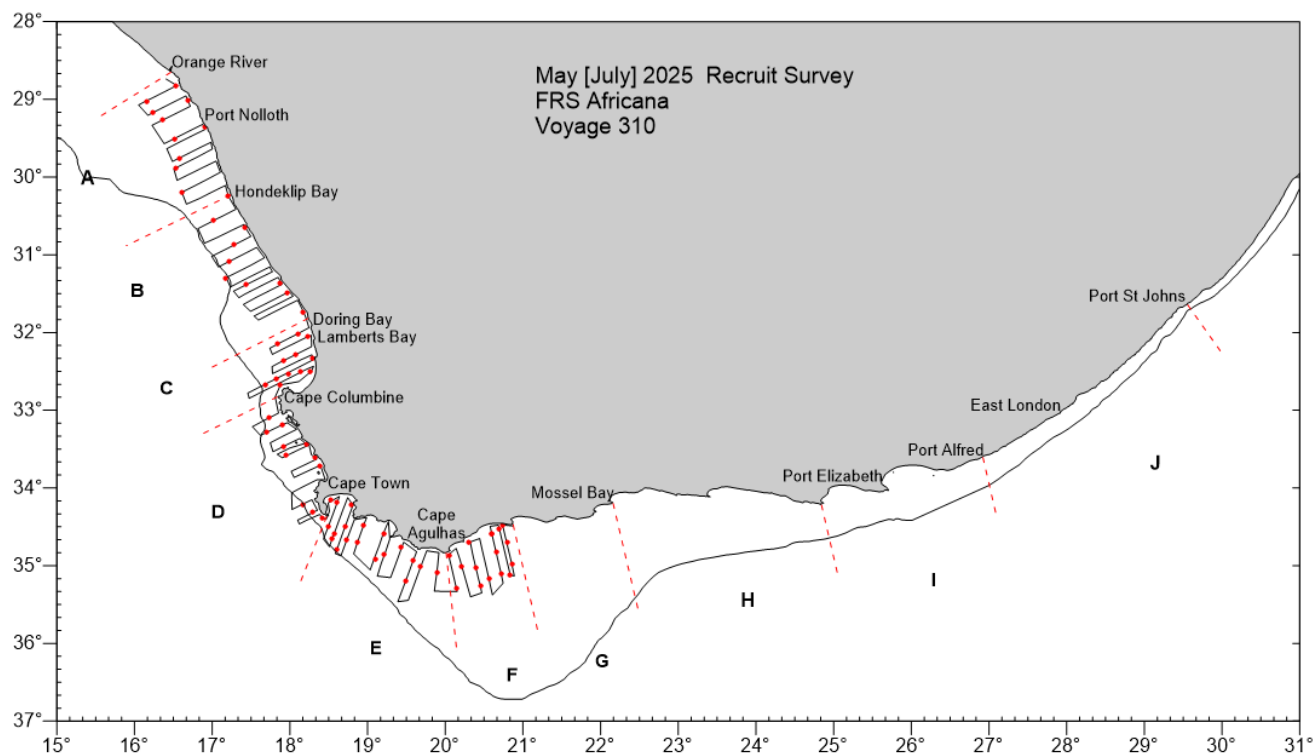


Figure 1: Ship's survey track, showing trawl positions between the Orange River and Cape Infanta.

Echo-sounder system

The SIMRAD EK80 scientific echo sounder system was operated continuously. A total of 312 acoustic intervals were logged along the 58 transects of the survey. All the intervals were used for computation of the biomass, except for offshore transits between sampling transects. Acoustic data were collected at 18, 38, 120, and 200 kHz in their RAW form. Echoview® Echolog software was used for echogram displays during the survey and data were processed using Echoview version 15.1. All the acoustic data were stored on logging computers and backed up on portable hard drives. Mean nautical area backscattering coefficients at 38 kHz were transformed to fish density for the calculation of fish biomass.

Given time constraints, an acoustic calibration was not conducted prior to the start of this survey and hence system settings from the March 2025 survey were used. Calibration of the system at that time, showed that the system and particularly the 38 kHz transducer had remained stable over the previous 18 months. That calibration required that a correction factor of 0.992 at 38 kHz be applied to the Nautical Area Scattering Coefficients obtained at 38 kHz (transmitting at 1000 W with a pulse duration of 1.025 ms) to account for slight differences in the gain and s_A correction factor. That correction has therefore also been applied to the current set of survey data.

Recruit cut-off lengths

The fish density estimates were also adjusted using the length frequencies from the trawl data to reflect the biomass and coefficient of variation (CV) of recruits only, following the calculation of recruit cut-off lengths using the previously agreed 20% length at gonad maturity stage approach based on biological sampling up to Cape Infanta. Recruit cut-off lengths of 8.5 cm caudal length for anchovy, 12 cm caudal length for sardine and 12 cm caudal length for redeye were obtained (see AppendixA).

Anchovy, sardine and round herring distribution patterns and size frequencies

1. Anchovy

On the west coast, anchovy recruits were distributed in patchy low density nearshore areas north of Port Nolloth and around Hondeklip Bay. A high density patch was found close inshore between Doring Bay and Cape Columbine and in False Bay (Figure 2). This patchy distribution is unusual for anchovy at this time of the year when the distribution typically tends to be an almost continuous near-shore band (Figure 2). No recruits were found east of False Bay as far as Cape Infanta. Patches of adult anchovy were located on the west coast between Doring Bay and St Helena Bay, and further south, off Cape Point, in False Bay and between Walker Bay and Cape Infanta, with highest densities off Doring Bay and just east of Cape Agulhas (Figure 3). The length frequency in the standard recruit survey area up to Cape Infanta was dominated by anchovy <9 cm (Figure 4); with the proportion of recruit fish (by number) being 67%, down from the 77% in 2024 but up from 59% in 2023. This is lower than typically observed (89% in 2022, 84% in 2021). The mean weight of recruits west of Cape Infanta of 3.56 g was higher compared to the 3.02 g of 2024 and the 3.33 g measured in 2023. Most of the anchovy were found in stratum C (Figure B1).

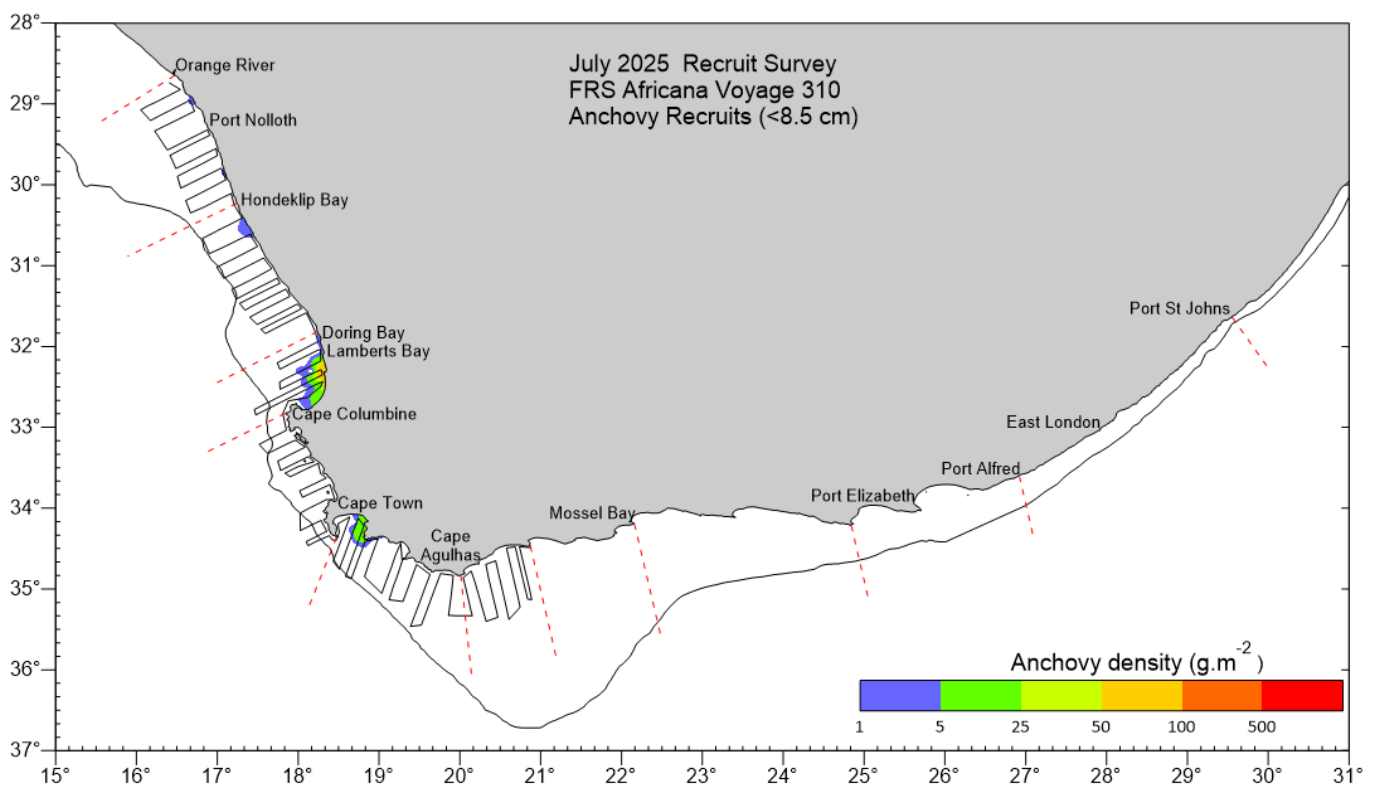


Figure 2: Distribution and relative abundance of anchovy recruits from the Orange River to Cape Infanta.

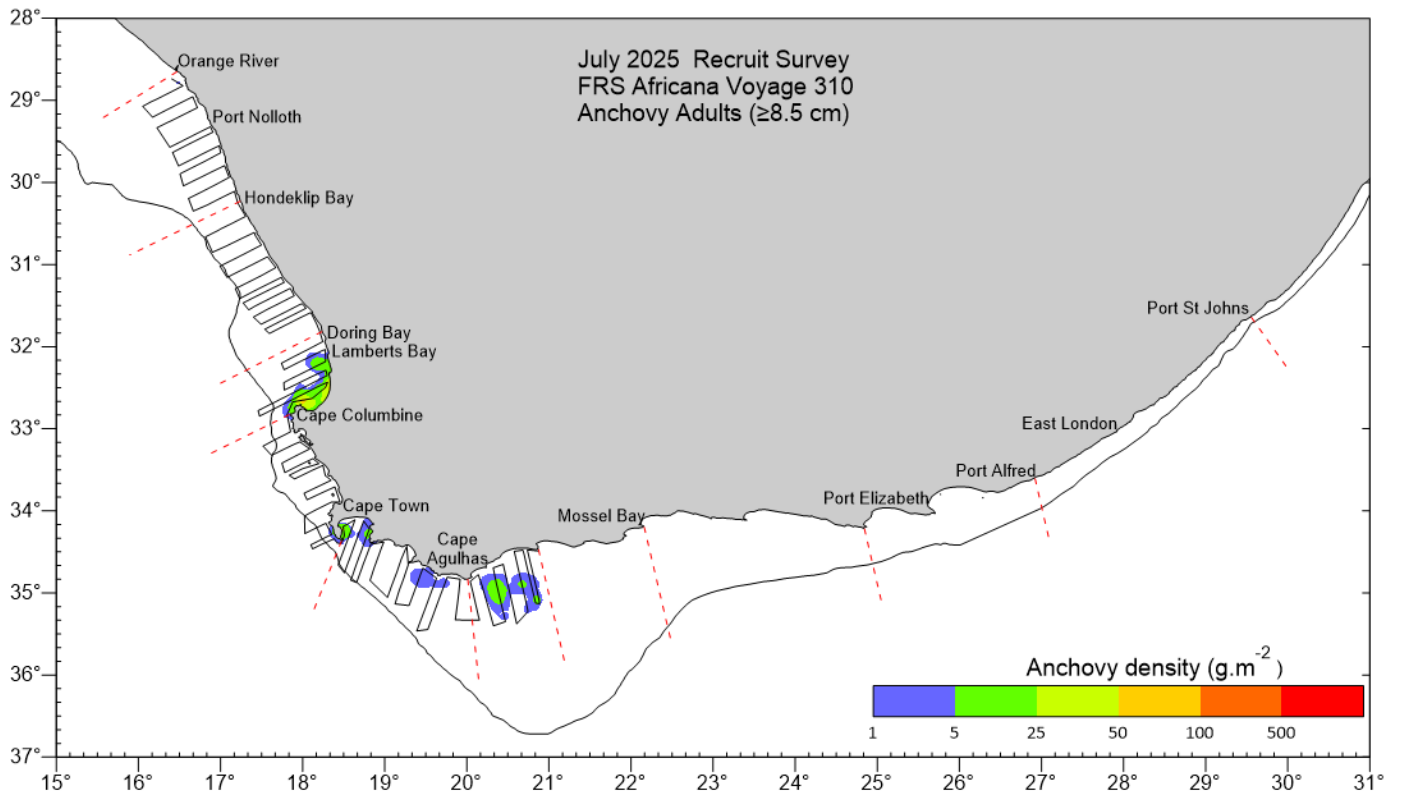


Figure 3: Distribution and relative abundance of adult anchovy from the Orange River to Cape Infanta.

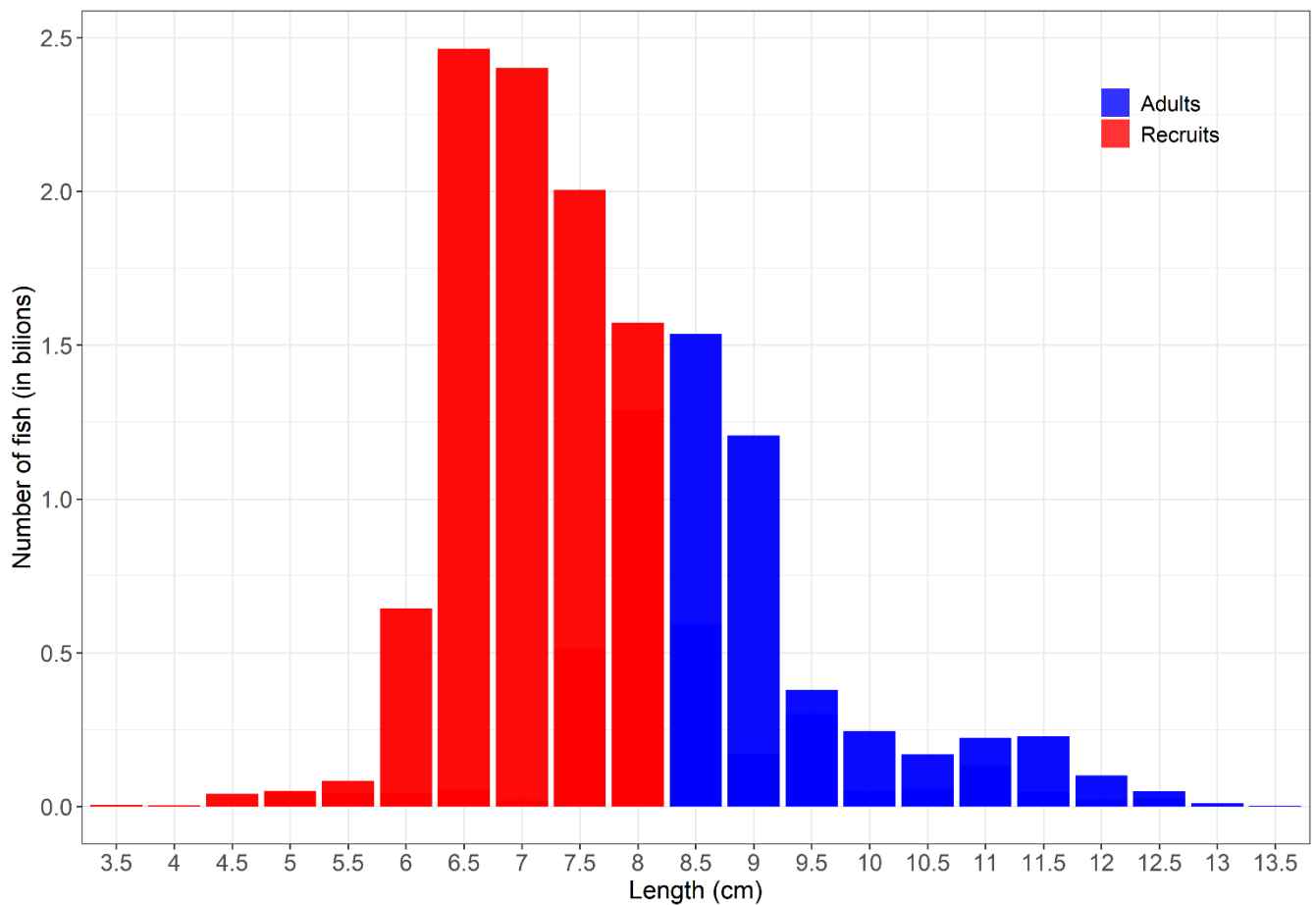


Figure 4: Acoustically weighted length frequency of anchovy recruits and adults from the Orange River to Cape Infanta.

2. Sardine

No sardine recruits were found north of St Helena Bay (Figure 5). A small, low-density patch of sardine was found inshore in St Helena Bay. A moderate to high density sardine recruit patch was found in False Bay, and in Walker Bay. A low-density patch of sardine recruits extended from the inshore to mid-shelf area between Cape Agulhas and Cape Infanta (Figure 5). The distribution of sardine adults (≥ 12 cm) encountered during the survey was slightly different to that of sardine recruits, with a few low densities of adult sardine found further offshore in the area between the Orange River and Doring Bay (Figure 6). However, adults co-occurred with recruits in False Bay, Walker Bay and in the area between Cape Agulhas and Cape Infanta but extended further offshore off Cape Agulhas. Sardine recruits west of Cape Agulhas were mostly bigger (> 10 cm) whereas those caught east of Cape Agulhas to Cape Infanta were mostly smaller than ~ 10 cm (Figure 7, Appendix B). The sardine length frequency up to Cape Infanta had many relatively big 10.5 and 11.5 cm fish (Figure 8). However, only 55% (up from 30% in 2023 but down from 64% in 2024) of sardines found west of Cape Infanta were classified as recruits, by number. This is lower than in 2021 and 2022 when 84 and 72 percent respectively, of sardine west of Cape Infanta were recruits. The mean weight of sardine recruits was much heavier (12 g) compared to the 3.89 g in 2024, 9.01 g in 2023, 5.19 g in 2021 and 6.99 g in 2022.

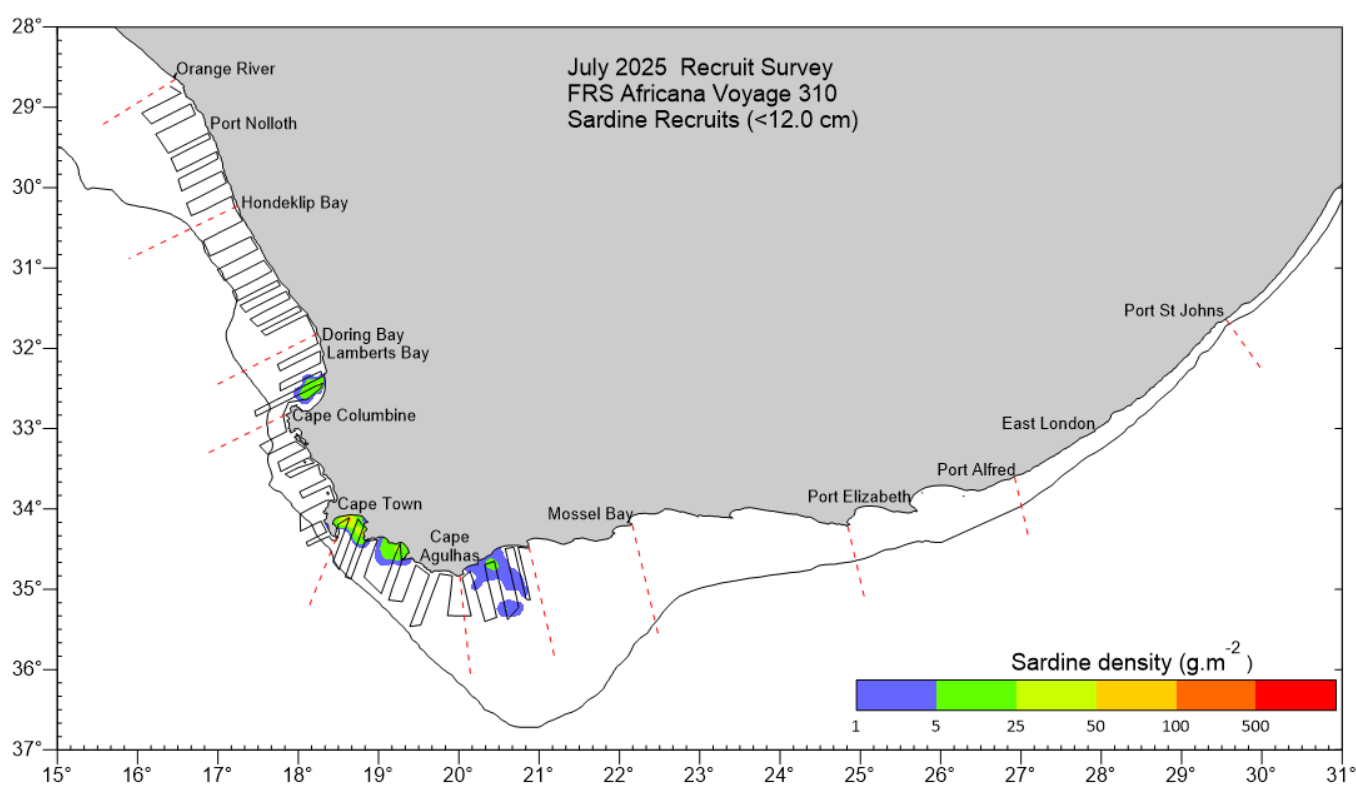


Figure 5: Distribution and relative abundance of sardine recruits from Orange River to Cape Infanta.

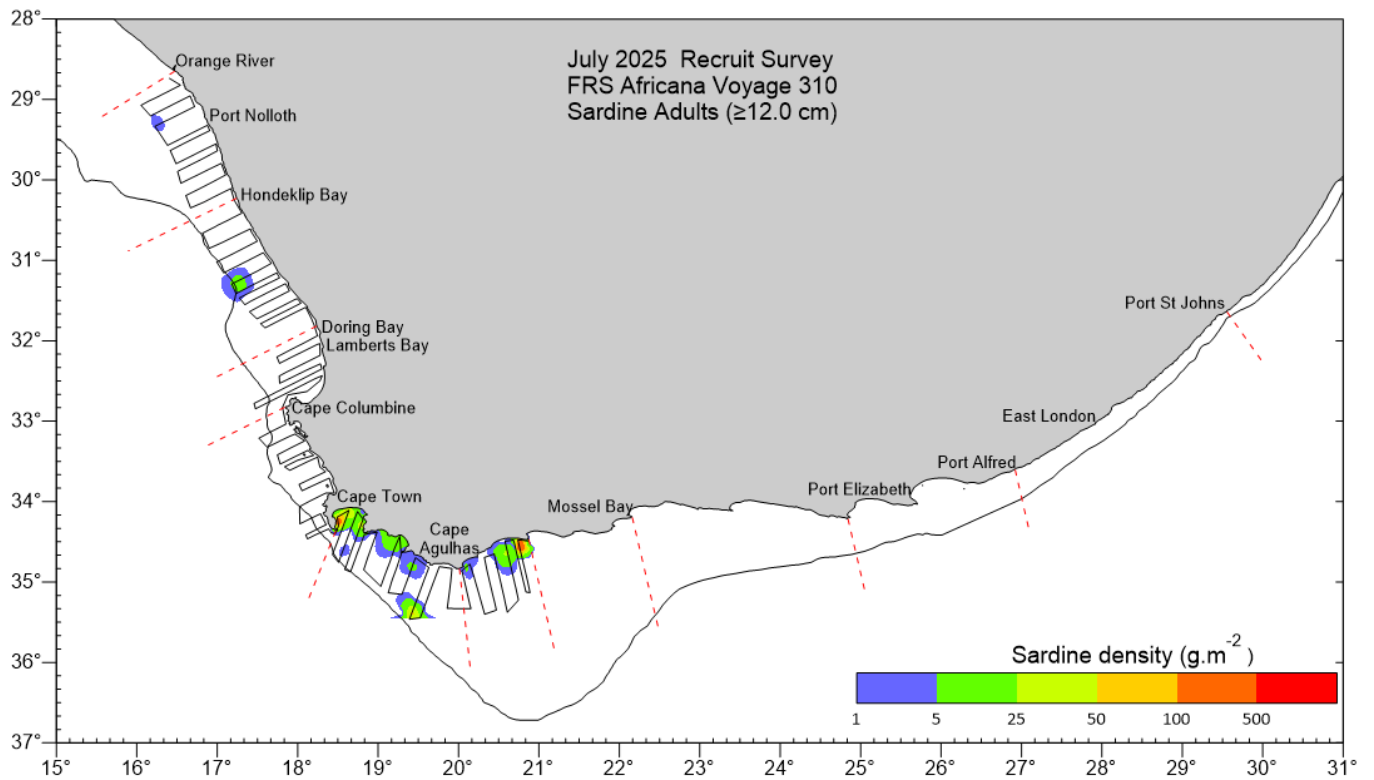


Figure 6: Distribution and relative abundance of sardine adults from Orange River to Cape Infanta.

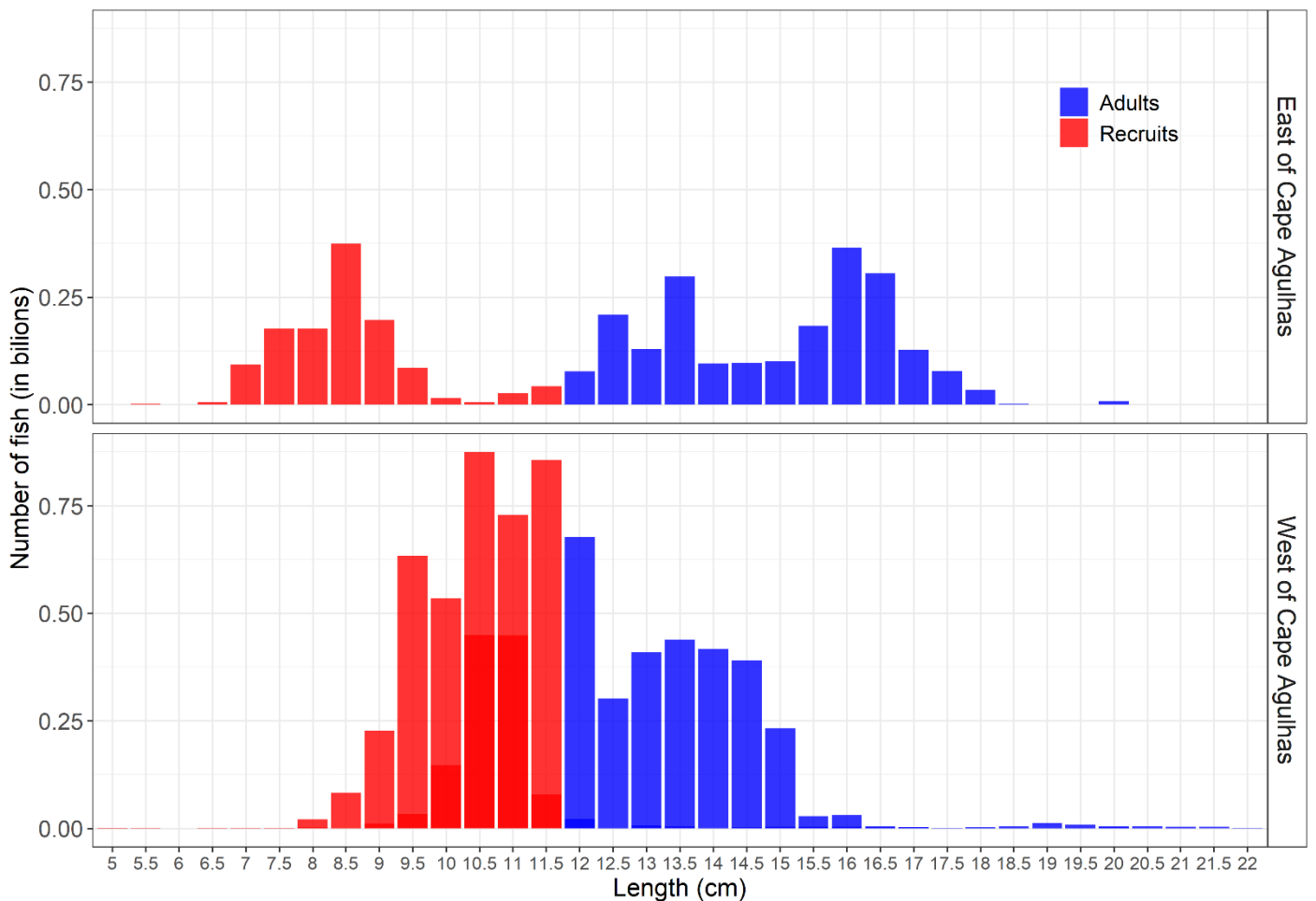


Figure 7: Acoustically weighted length frequency for sardine recruits and adults east and west of Cape Agulhas.

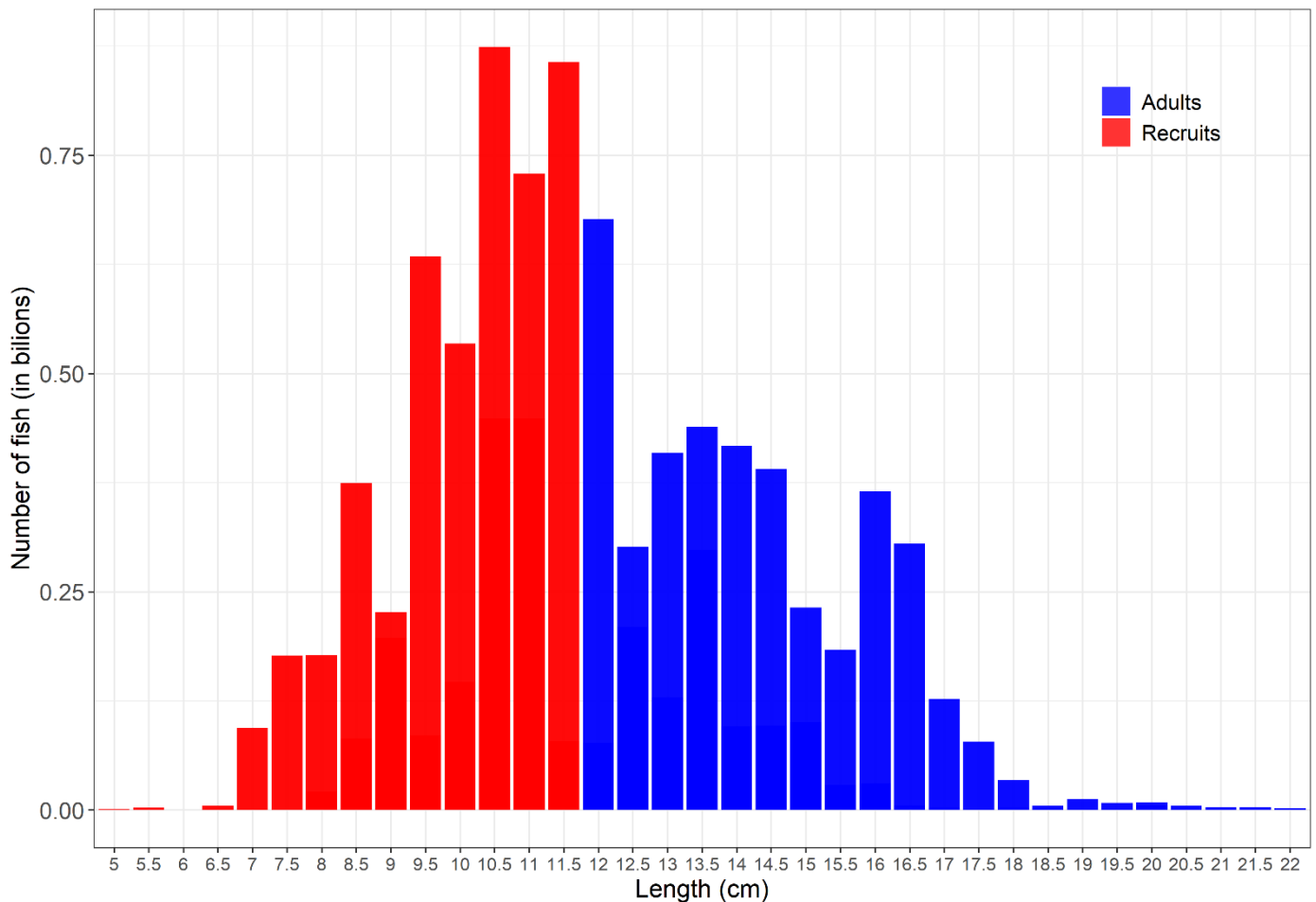


Figure 8: Acoustically weighted length frequency for sardine recruits and adults from Orange River to Cape Infanta.

3. Redeye

Most redeye recruits were found on the west coast with a high density patch in St Helena Bay (Figure 9) and a low density inshore strip close inshore between Port Nolloth and Doring Bay. Further south, low density area of redeye recruits were encountered stretching from inshore to the mid-shelf region between Cape Point and Cape Infanta (Figure 9). Redeye adults extended further offshore beyond the survey limits on the west coast in low to moderate densities. Off the south west coast, the distribution of redeye adults was almost continuous across the shelf between False Bay and Cape Infanta (Figure 10). Redeye recruits of 10 and 10.5 cm dominated the length frequency distribution for the area up to Cape Infanta (Figure 11).

Stratum-specific length frequency distributions are provided in Appendix B and distribution maps of chub mackerel, horse mackerel, lanternfish, and lightfish are provided in Appendix C.

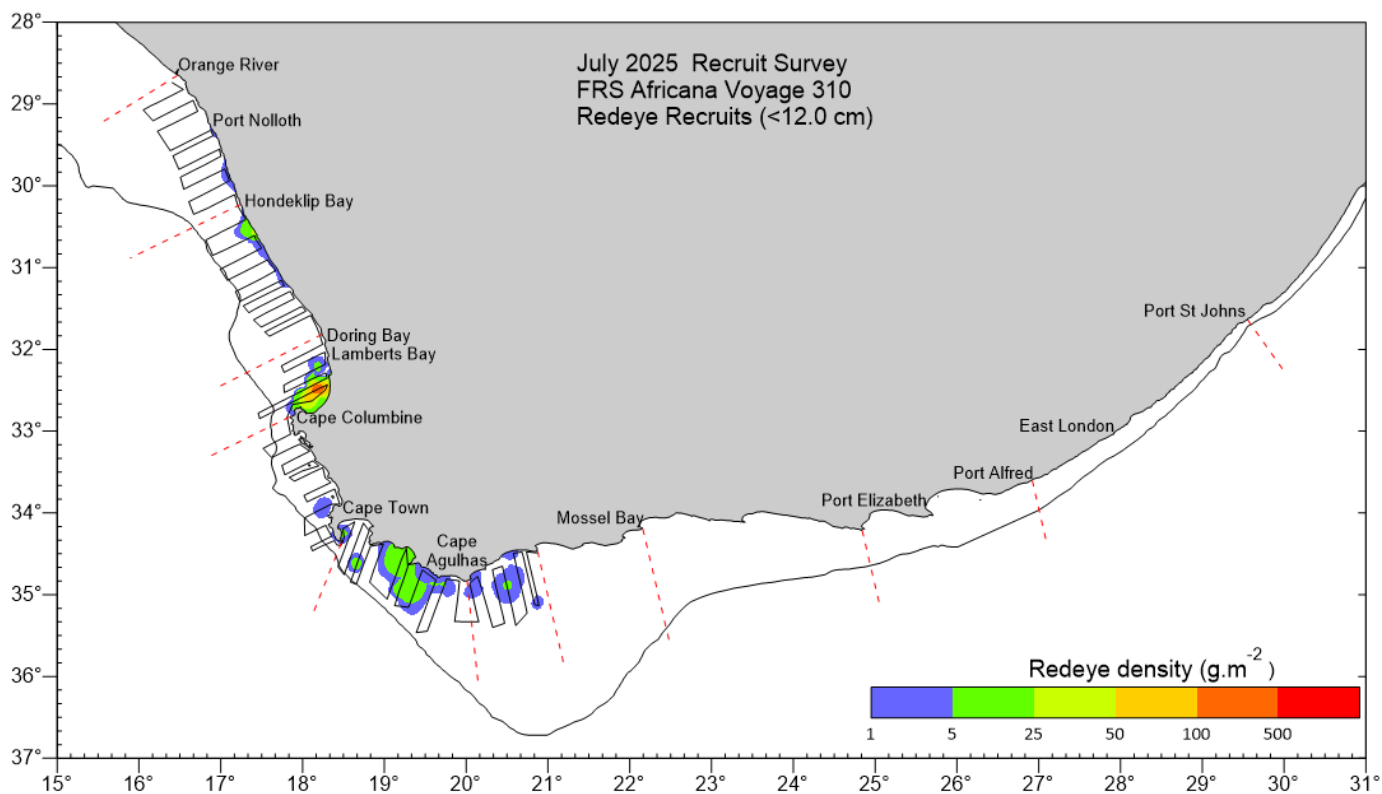


Figure 9: Distribution and relative abundance of redeye recruits from Orange River to Cape Infanta.

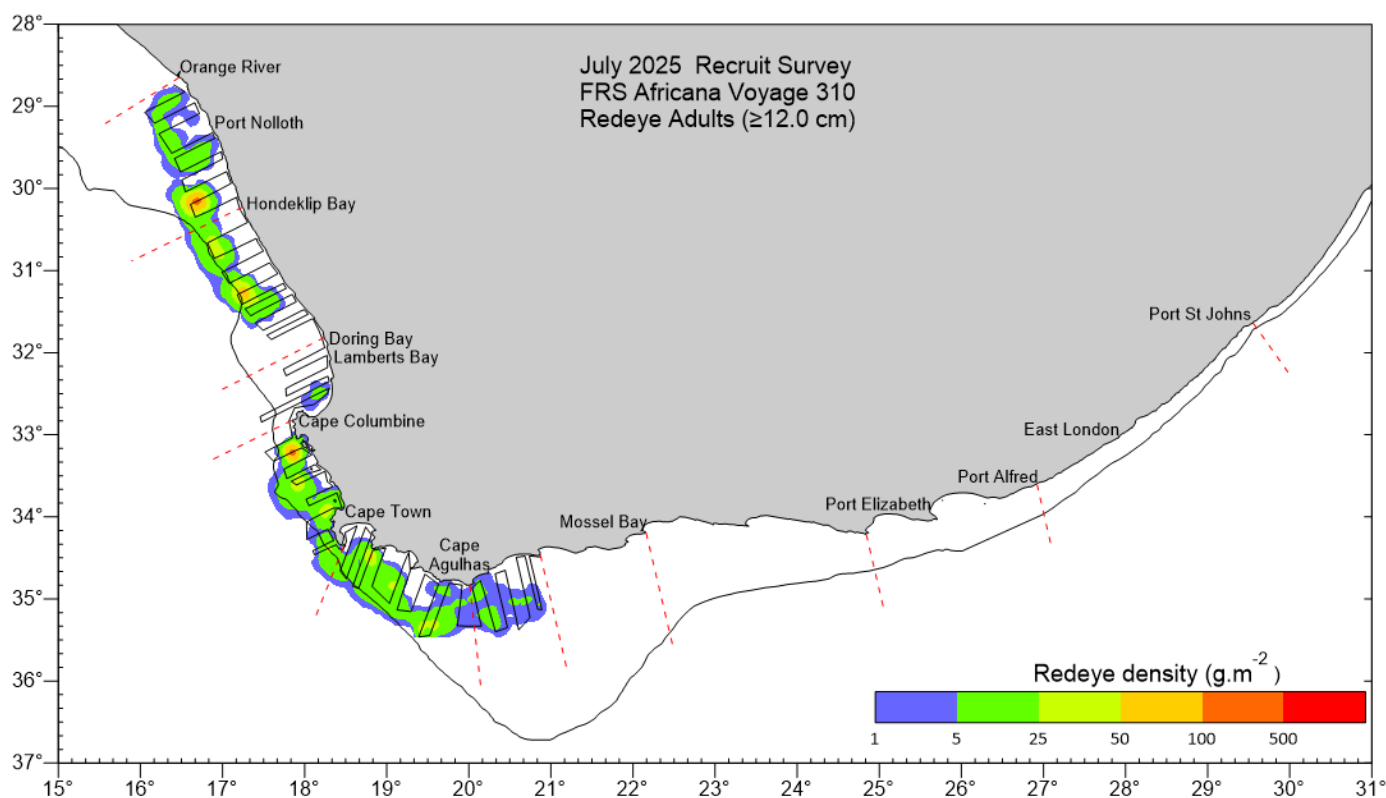


Figure 10: Distribution and relative abundance of redeye adults from Orange River to Cape Infanta.

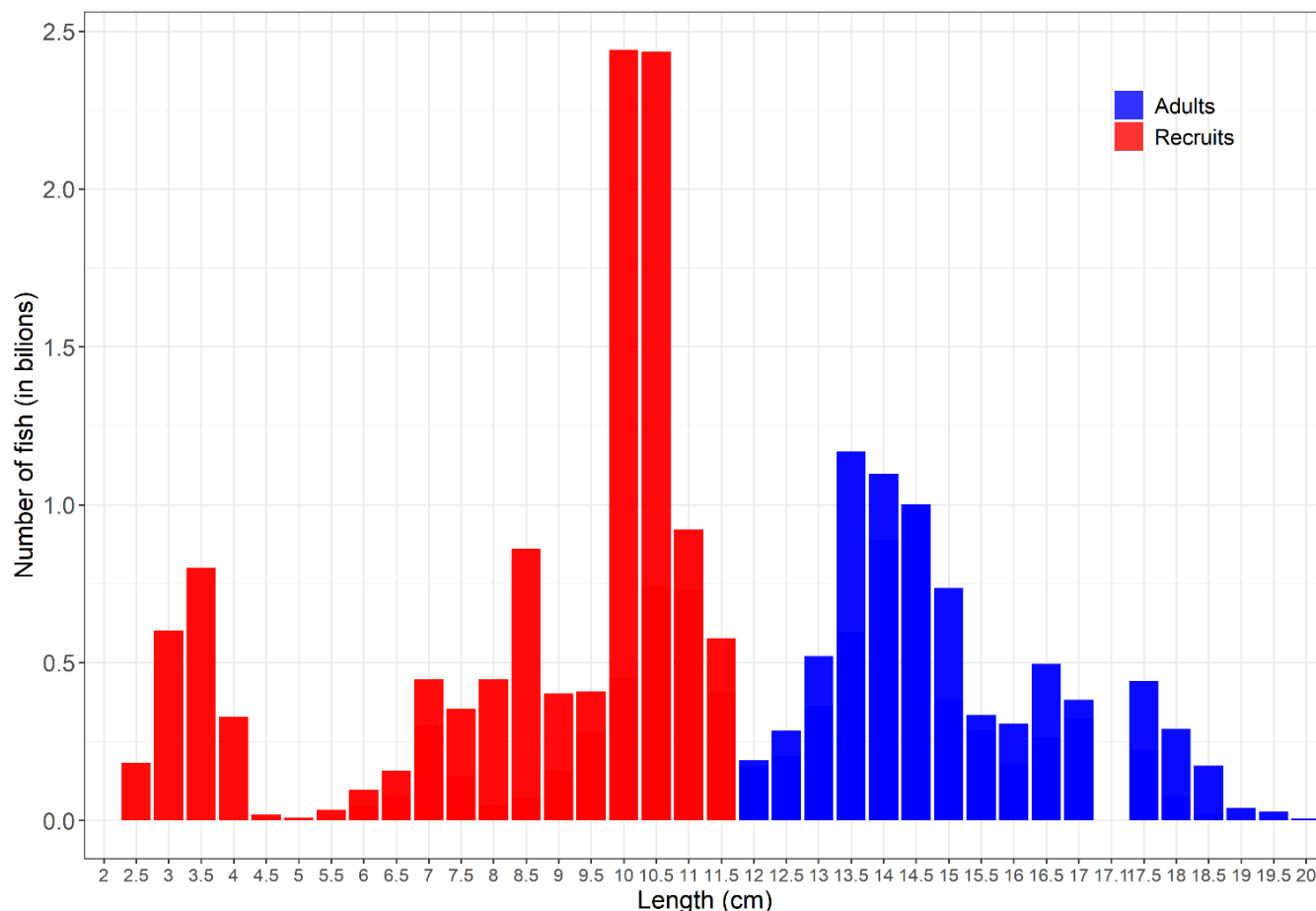


Figure 11: Acoustically weighted length frequency for redeye recruits and adults from the Orange River to Cape Infanta.

Stock size

Estimates of anchovy, sardine and redeye recruit and adult biomass per stratum and for the total area up to Cape Agulhas and Cape Infanta together with their CVs are shown in Tables 1-3.

The anchovy recruit biomass up to Cape Infanta of 43 000 tonnes is only a quarter of the 166 027 tonnes measured in 2024 and even lower than the 55 000t of 2023. This is the second lowest recruit biomass estimate on record and considerably lower than the long-term average of ~ 605 000 tonnes (Table 1 and 4). In terms of number, the recruit estimate of 11.6 billion fish is the lowest recorded and far below the 54.2 billion fish up to Cape Infanta in 2024. This recruitment represents only about 6% of the long-term average of 206 billion fish (Figure 12; Table 4) and is also considerably lower than the most recent 5- and 10-year average.

The sardine recruitment biomass to Cape Infanta amounts to 77 815 tonnes, which is about six times higher than the 13 000 tonnes estimated in 2024 and the highest in terms of biomass since 2015. Whereas this estimate is still below the long-term average of ~ 103 000 tonnes, it is more than double the most recent 5- and 10-year average (Table 2 and 4). The recruit estimate up to Cape Infanta in terms of number of fish (6.34 billion) is 63% higher than the 3.58 billion fish measured in 2024 and although still well below the long-term average of 10.27 billion fish (Figure 12; Table 4) it is > 40% higher than both the recent 5- and 10-year averages. The sardine recruit estimates of 5.135 billion fish and 69 165 tonnes estimated for the area to the west of Cape Agulhas remains below the long-term average for this area, but are above the most recent averages. The total biomass of sardine (including the adults) up to Infanta, occurring within the

inshore recruit survey area was ~250 000 tonnes, of which ~150 000 tonnes was located to the west of Cape Agulhas.

Table 1. Anchovy recruit (<8.5 cm) and adult biomass estimates (tonnes) and estimated CV during the survey per stratum and for the standard survey area up to Cape Infanta. Also shown is the number of recruits.

Stratum	Recruit biomass	CV	Recruits (x10 ⁰⁹)	Adult Biomass	Total biomass	CV
A	384	0.490	0.181	384	768	0.325
B	489	0.684	0.264	35	524	0.688
C	32 404	0.571	8.936	21 115	53 518	0.334
D	4	0.452	0.009	66	69	0.454
E	9 421	0.880	2.106	8 545	17 965	0.579
F	362	0.560	0.101	16 105	16 467	0.386
Total Cape Agulhas	42 701	0.475	11.496	30 144	72 845	0.284
Total Cape Infanta	43 063	0.471	11.597	46 249	89 312	0.242

Table 2. Sardine recruit (<12 cm) and adult biomass estimates (tonnes) and estimated CV during the survey per stratum and for the survey area up to Cape Agulhas and Cape Infanta. Also shown is the number of recruits.

Stratum	Recruit biomass	CV	Recruits (x10 ⁰⁹)	Adult Biomass	Total biomass	CV
A	4	0.654	0.001	1 209	1 213	0.684
B	45	0.720	0.008	5 386	5 431	0.637
C	18 443	0.933	1.169	507	18 951	0.933
D	2	0.598	0.000	1 130	1 131	0.346
E	50 671	0.416	3.957	79 815	130 485	0.304
F	8 650	0.312	1.206	83 954	92 604	0.621
Total Cape Agulhas	69 165	0.393	5.135	88 047	157 212	0.277
Total Cape Infanta	77 815	0.351	6.341	172 001	249 816	0.289

The redeye recruit biomass estimate up to Infanta of 156 000 tonnes is slightly higher (+8%) than the ~144 000 tonnes estimated in 2024, and much higher than the long-term average of around 74 000 tonnes (Table 3 and 4). The number of recruits (15.9 billion fish) is also higher than that measured in 2024 (13.88 billion fish) and above the 12.2 billion fish long-term average (Figure 12; Table 4).

Table 3. Redeye recruit (<12 cm) and adult biomass estimates (tonnes) and estimated CV during the survey per stratum and for the survey area up to Cape Agulhas and Cape Infanta. Also shown is the number of recruits.

Stratum	Recruit biomass	CV	Recruits (x10 ⁰⁹)	Adult Biomass	Total biomass	CV
A	645	0.477	0.198	106 944	107 589	0.429
B	2 650	0.744	0.616	111 023	113 673	0.268
C	105 303	0.820	9.473	6 607	111 909	0.826
D	3 045	0.343	2.123	142 555	145 600	0.358
E	35 986	0.303	2.879	142 011	177 997	0.150
F	8 408	0.242	0.614	24 843	33 251	0.175
Total Cape Agulhas	147 630	0.590	15.288	509 139	656 769	0.187
Total Cape Infanta	156 038	0.558	15.902	533 981	690 019	0.178

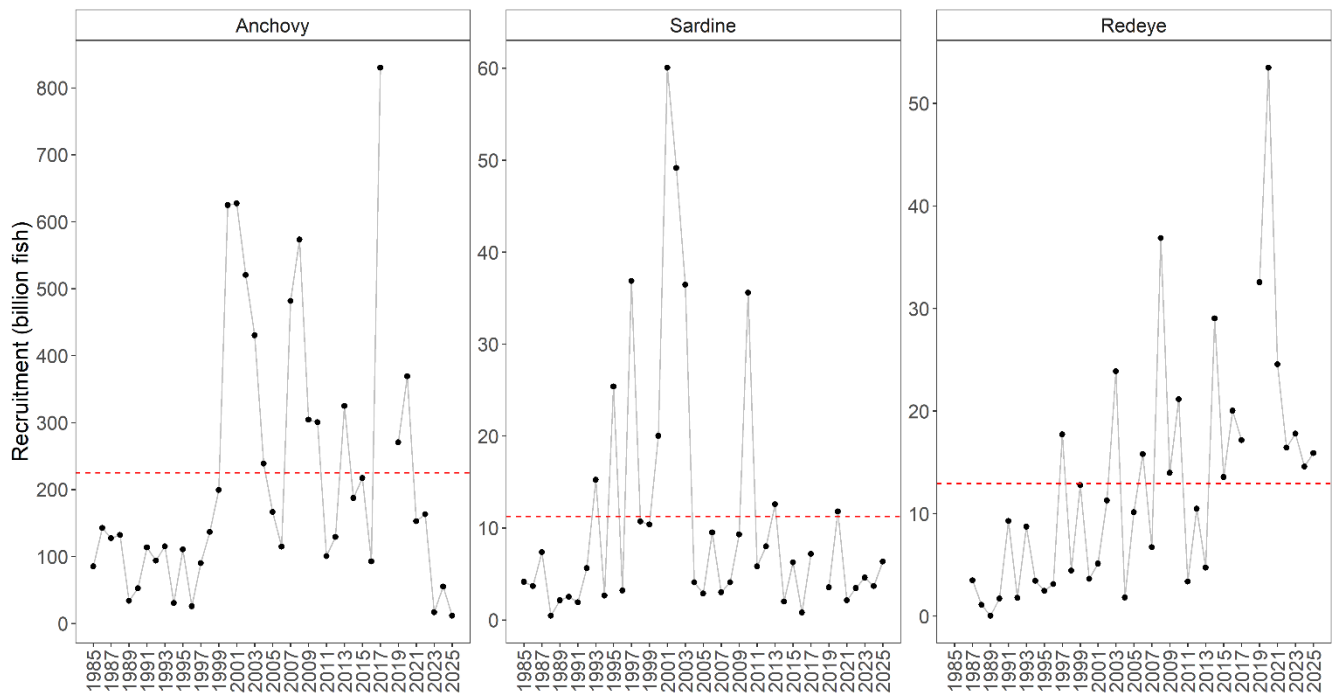


Figure 12: Number of recruits estimated in May/June recruitment surveys west of Cape Infanta since 1985 for anchovy (left), sardine (middle) and redeye/round herring (right). The average from 1985-2025 is shown as a dashed horizontal red line in each case.

Table 4. Anchovy, sardine, and redeye recruitment estimates up to Cape Infanta. Also shown are the long-term average (1985-2024), the recent 10-year (2014-2024) and 5-year averages (2020-2024).

	Anchovy recruitment			Sardine recruitment			Redeye recruitment		
	Numbers	Biomass	CV (%)	Numbers	Biomass	CV (%)	Number	Biomass	CV (%)
1985	49.8	117 234	40.3	4.0	37 061	56.6			
1986	128.7	521 313	17.5	3.2	18 434	48.0			
1987	40.4	107 804	39.9	5.8	56 926	57.1	3.9	19 950	44.2
1988	47.8	101 985	21.1	0.2	1 357	50.1	1.2	10 022	42.5
1989	12.3	25 917	47.9	0.6	7 414	54.8	0.0	205	51.8
1990	45.6	133 232	26.9	2.1	16 888	74.8	1.8	8 700	54.9
1991	71.6	239 361	20.3	1.8	19 169	29.3	7.8	46 273	49.3
1992	55.7	165 582	20.0	5.1	49 686	40.7	1.8	15 159	26.3
1993	73.9	139 006	20.7	14.6	91 023	38.3	7.5	27 422	37.6
1994	21.7	79 797	19.8	1.4	14 150	35.2	3.4	21 243	26.5
1995	105.8	353 262	17.1	25.1	182 228	36.7	2.3	15 373	30.8
1996	22.3	51 658	24.8	2.3	35 458	37.2	3.0	28 401	43.0
1997	76.5	305 468	18.7	34.2	275 134	39.3	18.6	93 432	20.3
1998	128.7	381 715	15.6	9.4	104 144	35.4	4.0	25 794	25.5
1999	170.3	625 411	18.6	8.2	128 090	43.3	13.3	56 824	19.9
2000	623.0	2 465 941	16.8	16.3	204 701	48.0	3.1	21 647	27.9
2001	598.5	1 769 930	13.9	60.0	558 798	28.7	3.4	27 808	56.0
2002	479.8	1 206 766	12.4	47.1	507 142	18.8	10.0	82 968	83.6
2003	430.3	1 396 638	18.9	32.4	452 647	21.3	19.5	128 912	34.7
2004	201.7	729 333	21.6	3.9	26 003	34.2	1.8	13 811	23.7
2005	126.7	253 228	22.3	2.9	16 437	35.2	7.4	72 232	28.2
2006	114.5	242 583	15.9	9.3	41 125	42.9	15.5	72 580	18.0
2007	480.1	1 309 951	19.7	2.9	32 777	34.3	6.7	77 008	26.4
2008	573.4	1 507 693	20.2	4.0	21 973	26.1	33.9	192 824	53.9
2009	304.4	844 721	23.1	6.6	57 236	74.8	13.9	72 790	34.2
2010	259.1	732 518	30.2	32.3	479 609	47.3	21.1	105 050	39.0
2011	98.1	246 548	28.4	5.1	58 282	46.0	2.3	17 443	33.2
2012	82.7	246 999	23.1	7.9	96 042	30.6	8.6	23 398	37.3
2013	291.3	803 100	19.5	11.7	95 255	41.3	4.5	27 958	23.3
2014	187.3	595 586	33.6	2.0	24 061	63.1	26.3	95 582	35.7
2015	183.2	588 802	18.1	5.4	116 027	57.4	13.2	106 760	17.2
2016	76.3	253 890	17.9	0.9	14 539	41.5	19.4	211 084	39.8
2017	766.5	2 211 337	14.3	6.8	18 565	62.8	17.2	64 158	24.4
2019	256.2	507 412	28.2	3.4	17 330	78.8	32.5	165 605	19.6
2020	451.4	1 464 887	18.7	7.9	54 294	36.6	51.5	283 273	19.5
2021	145.4	386 508	17.0	2.1	9 619	41.2	24.6	115 823	21.0
2022	162.9	246 201	20.1	3.5	26 686	55.5	16.0	81 524	18.8
2023	16.5	54 590	22.6	4.6	31 363	43.1	17.2	165 148	24.6
2024	54.2	166 027	30.4	3.6	13 009	38.6	13.9	144 061	18.3
2025	11.6	43 062	47.1	6.3	77 815	35.1	15.9	156 038	55.8
Average	205.5	604 613	22.5	10.3	102 838	44.2	12.2	74 007	33.3
(10 yr)	166.1	463 642	21.7	4.3	26 994	43.0	24.6	157 966	20.5
(5 yr)	230.0	647 524	22.1	4.0	32 549	51.9	23.2	143 302	23.9

Table 5. Sardine recruitment estimates between the Orange River mouth and Cape Agulhas and for the area between Cape Agulhas and Cape St Francis as well as for the total area to Cape St Francis. Also shown are the long-term (1985-2024), the recent 10-year (2014-2024) and recent 5-year averages (2012-2024).

Sardine recruitment west and east of Cape Agulhas									
	Orange River to Cape Agulhas			Cape Agulhas to Cape St Francis			Orange River to Cape St Francis		
	Numbers	Biomass	CV (%)	Numbers	Biomass	CV (%)	Number	Biomass	CV (%)
1985	4.02	36 997	56.7						
1986	3.21	18 394	48.0						
1987	5.61	56 803	57.1						
1988	0.23	1 354	50.1						
1989	0.58	7 398	54.9						
1990	2.12	16 852	74.8						
1991	1.82	19 169	29.3						
1992	5.10	49 634	40.7						
1993	14.61	91 016	38.3						
1994	1.42	14 117	35.3	0.03	549	46.70	1.44	14 666	0.34
1995	25.09	182 118	36.8	0.2	2 215	49.10	25.29	184 334	0.363
1996	2.26	27 767	41.7						
1997	33.97	270 761	39.9						
1998	9.73	103 220	35.7	0.19	3 083	55.80	9.92	106 303	0.347
1999	5.22	63 666	30.5	3.43	76 230	68.70	8.65	139 896	0.399
2000	18.77	203 890	48.2	3.65	95 816	43.80	22.42	299 706	0.356
2001	59.96	557 280	28.8	0.07	1 521	99.50	60.04	558 801	0.287
2002	44.28	424 075	20.6	4.04	117 887	45.00	48.32	541 962	0.188
2003	32.82	374 569	24.3	2.66	84 233	38.20	35.48	458 802	0.211
2004	4.07	25 617	34.7	0.6	4 863	67.80	4.67	30 480	0.311
2005	2.67	14 216	40.0	0.77	11 942	37.60	3.44	26 157	0.277
2006	9.22	40 646	43.4	4.65	80 249	52.30	13.87	120 895	0.377
2007	2.93	32 638	34.5	1.34	19 356	88.60	4.27	51 994	0.394
2008	3.99	21 791	26.3						
2009	9.15	55 635	76.9	3.78	65 491	98.70	12.93	121 126	0.64
2010	35.12	470 373	48.1	0.8	16 018	66.40	35.92	486 390	0.466
2011	5.41	55 398	45.0						
2012	6.80	77 681	36.5						
2013	11.69	95 229	41.3						
2014	1.93	23 634	64.2	1.04	22 327	69.70	2.97	45 961	0.473
2015	0.54	10 887	34.2	5.37	115 900	57.60	5.91	126 786	0.528
2016	0.88	14 379	42.0	2.26	61 053	89.40	3.14	75 432	0.728
2017	6.62	16 365	70.6	0.82	9 471	42.90	7.44	25 836	0.474
2018									
2019	2.80	9 848	112.3						
2020	7.10	45 669	42.6						
2021	2.10	9 419	42.1	0.07	669	42.70	2.17	10 088	0.394
2022	3.00	22 400	65.0	2.89	31 784	40.70	5.89	54 184	0.36
2023	3.98	25 489	48.5	0.66	6 138	88.40	4.64	31 627	0.427
2024	3.36	10 582	46.6	4.93	33 941	52.40	8.3	44 523	0.414
2025	6.34	77 815	35.1						
Average	10.11	92 230	45.8	2.01	39 124	61.0	14.87	161 634	0.4
(10 yr)	3.91	22 712	49.0	2.14	18 133	56.1	5.25	35 106	0.4
(5 yr)	3.23	18 867	56.8	2.26	35 160	60.5	5.06	51 805	0.5

Conclusions

The survey appears to have covered the major distribution of pelagic recruits, which were mostly confined to inshore areas shallower than 50 m and in many cases found along inshore transits between transect lines. Adult biomasses referred to here are an underestimate of true adult biomass as the survey was designed to target recruits and would have missed adults distributed further offshore. The decrease in anchovy recruitment in the standard survey area since 2023 is confirmed and expected given the recent lowest biomass estimates in 2024 and in March/April 2025. Sardine recruitment measured west of Cape Agulhas is still below the long-term average but but substantially higher than the recent 5- and 10-year averages. Unfortunately the survey area between Cape Agulhas and Cape St Francis could not be surveyed due to a lack of time but encouraging signs of both recruits and adults were seen in the area between Cape Agulhas and Cape Infanta. Round herring recruitment has been relatively stable since 2022 and remains above the long-term average.

Appendix A: Maturity ogives based on macroscopic maturity stages 2-7

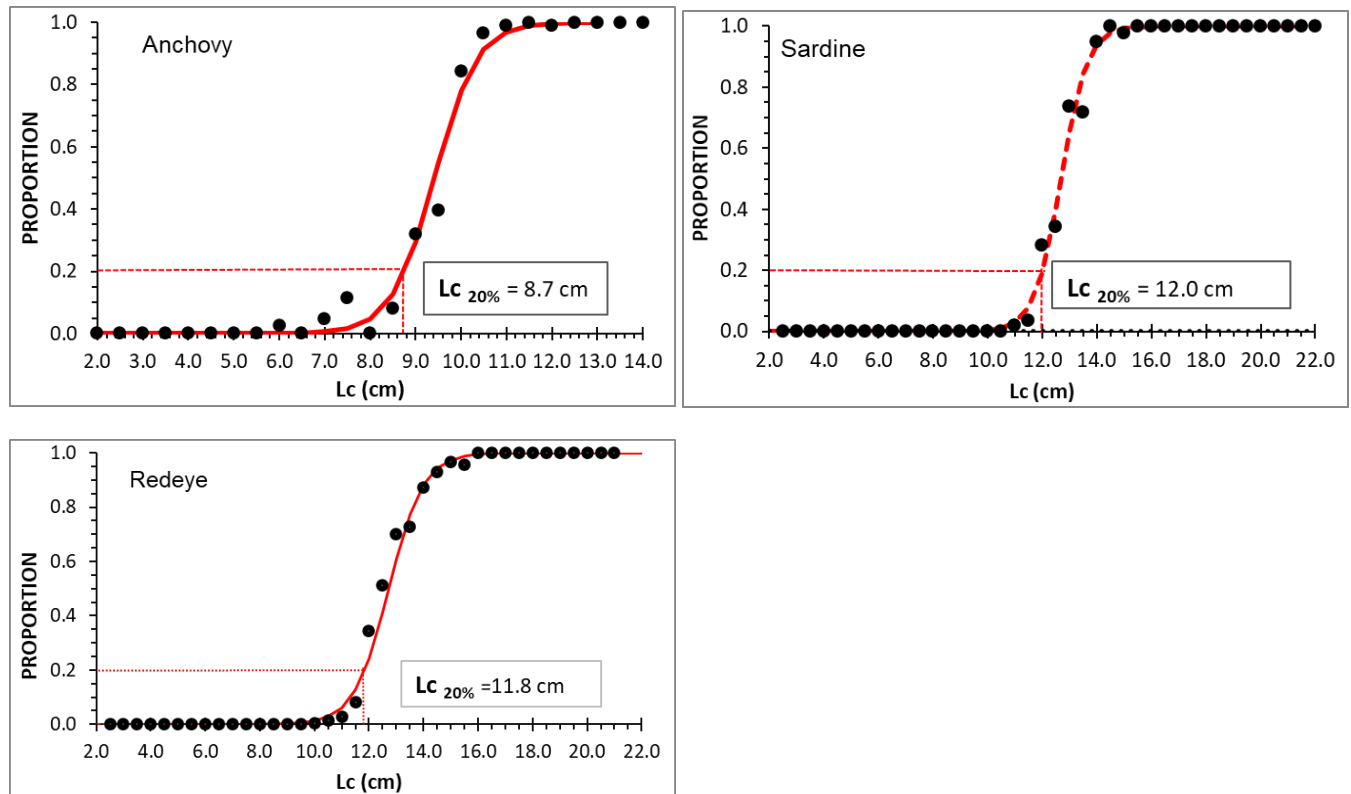


Figure A1. Maturity ogives of the three small pelagic fish. Red horizontal lines at 0.2 represent the 20% length at maturity.

Appendix B: Length frequencies

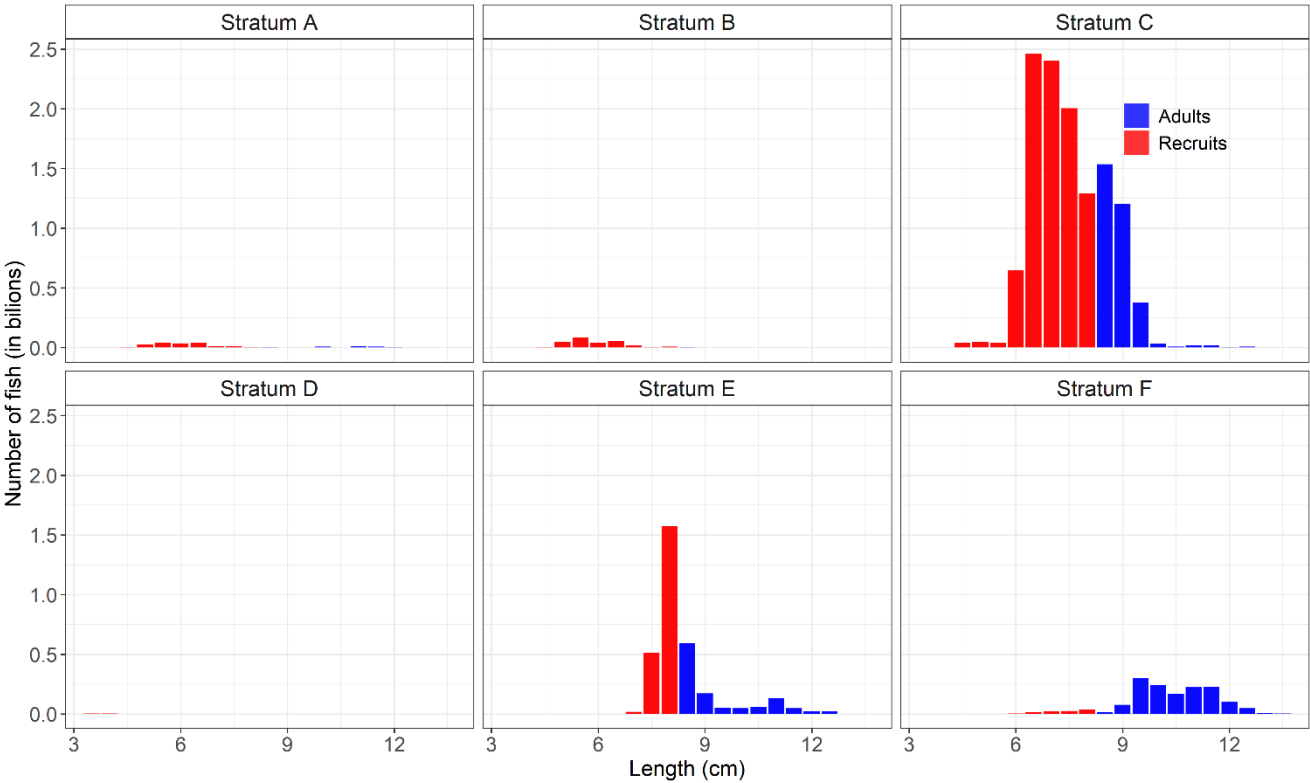


Figure B1: Anchovy length frequencies per stratum.

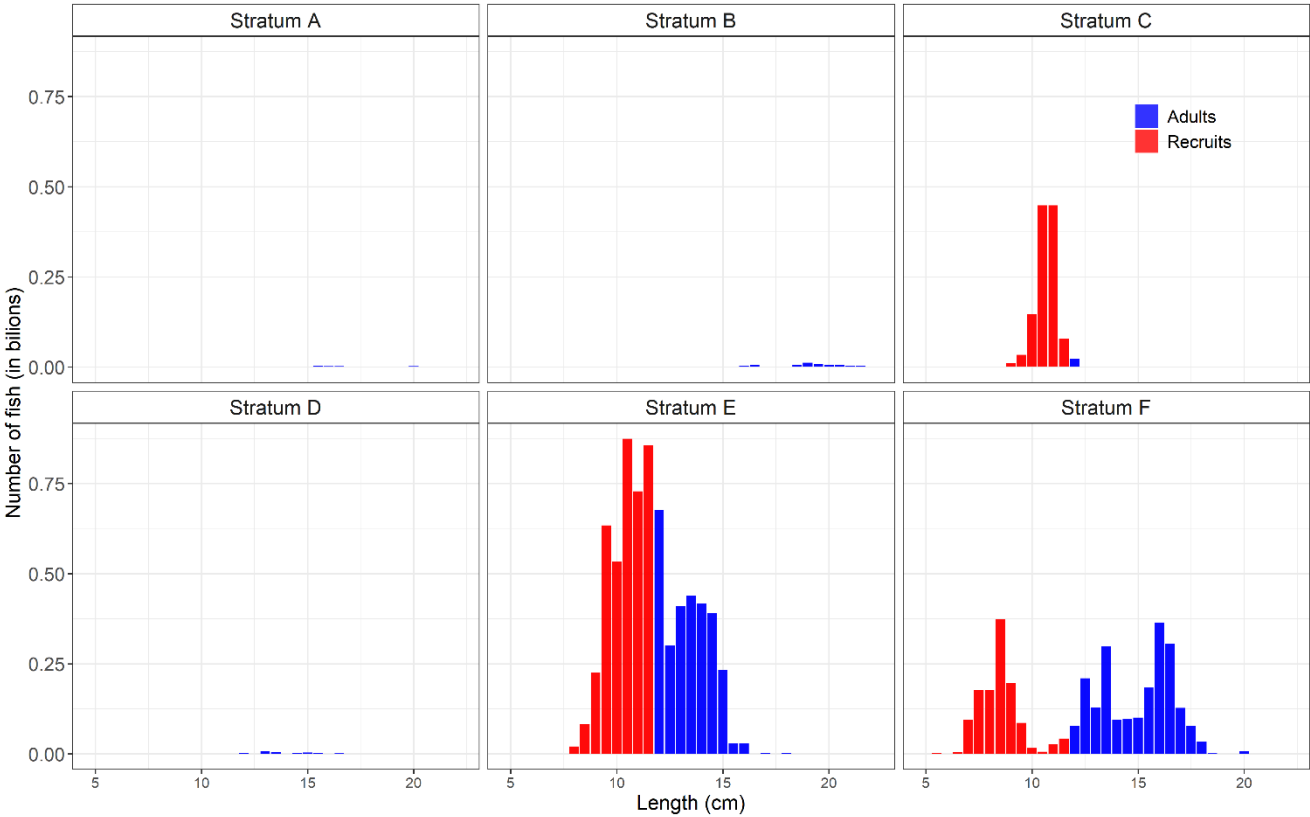


Figure B2: Sardine length frequencies per stratum.

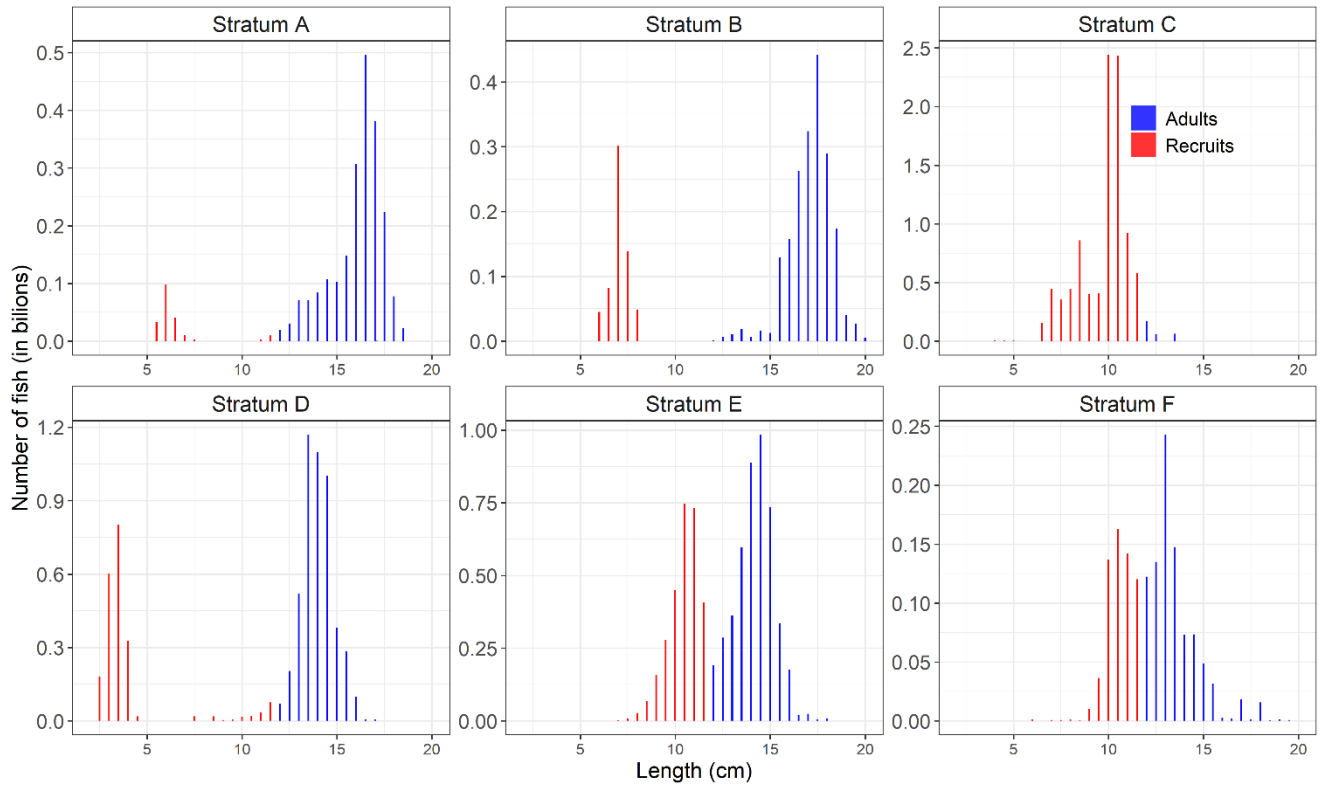


Figure B3: Redeye length frequencies per stratum.

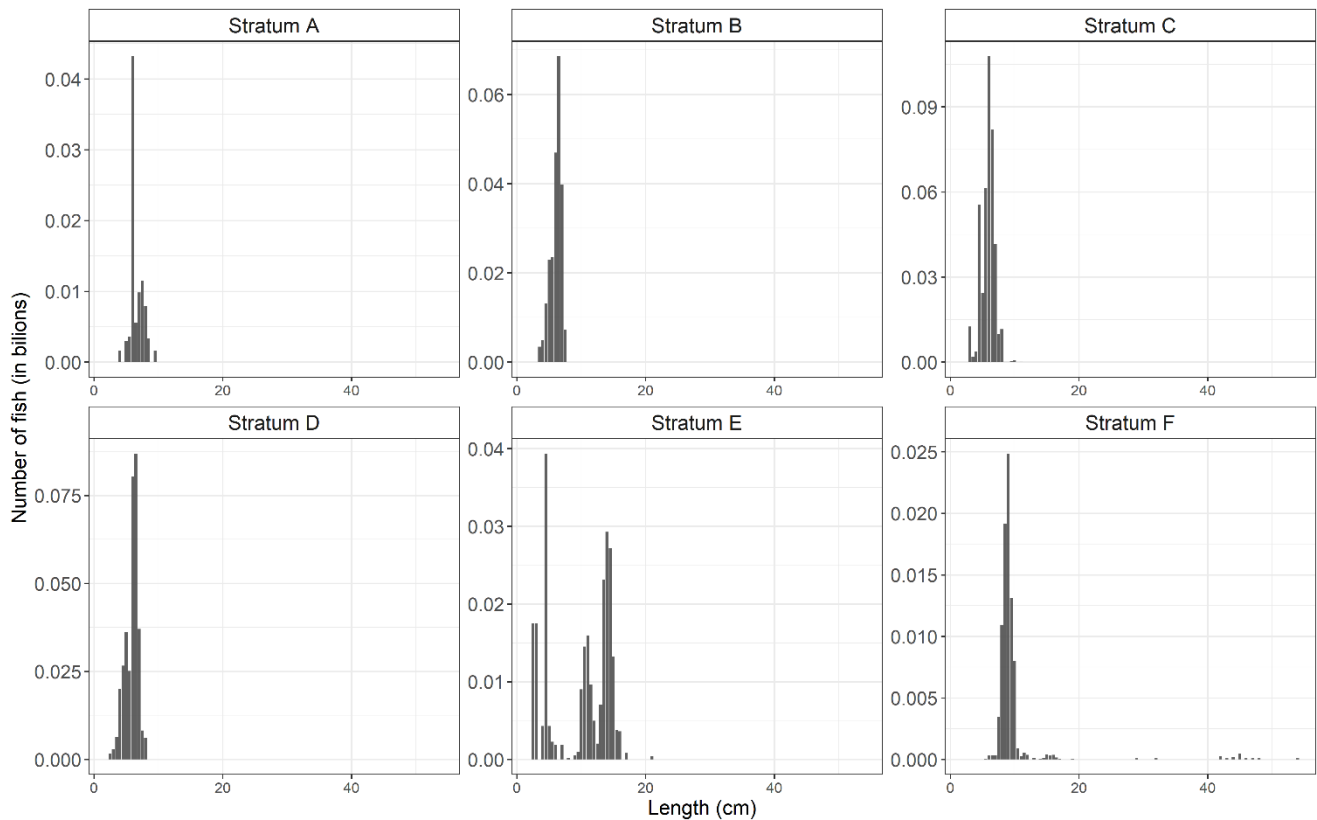


Figure B4: Horse mackerel length frequencies per stratum.

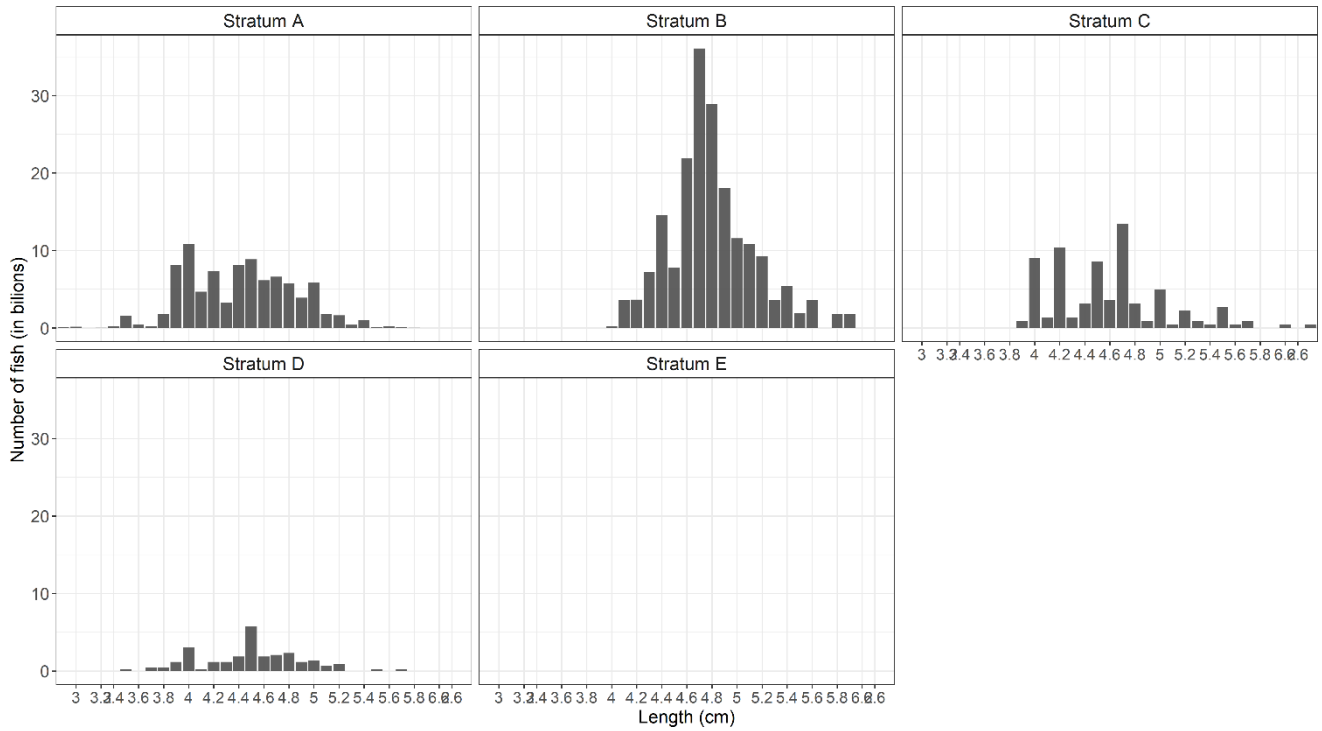


Figure B5: Length frequencies of light fish per stratum. No light fish were found east of Cape Agulhas.

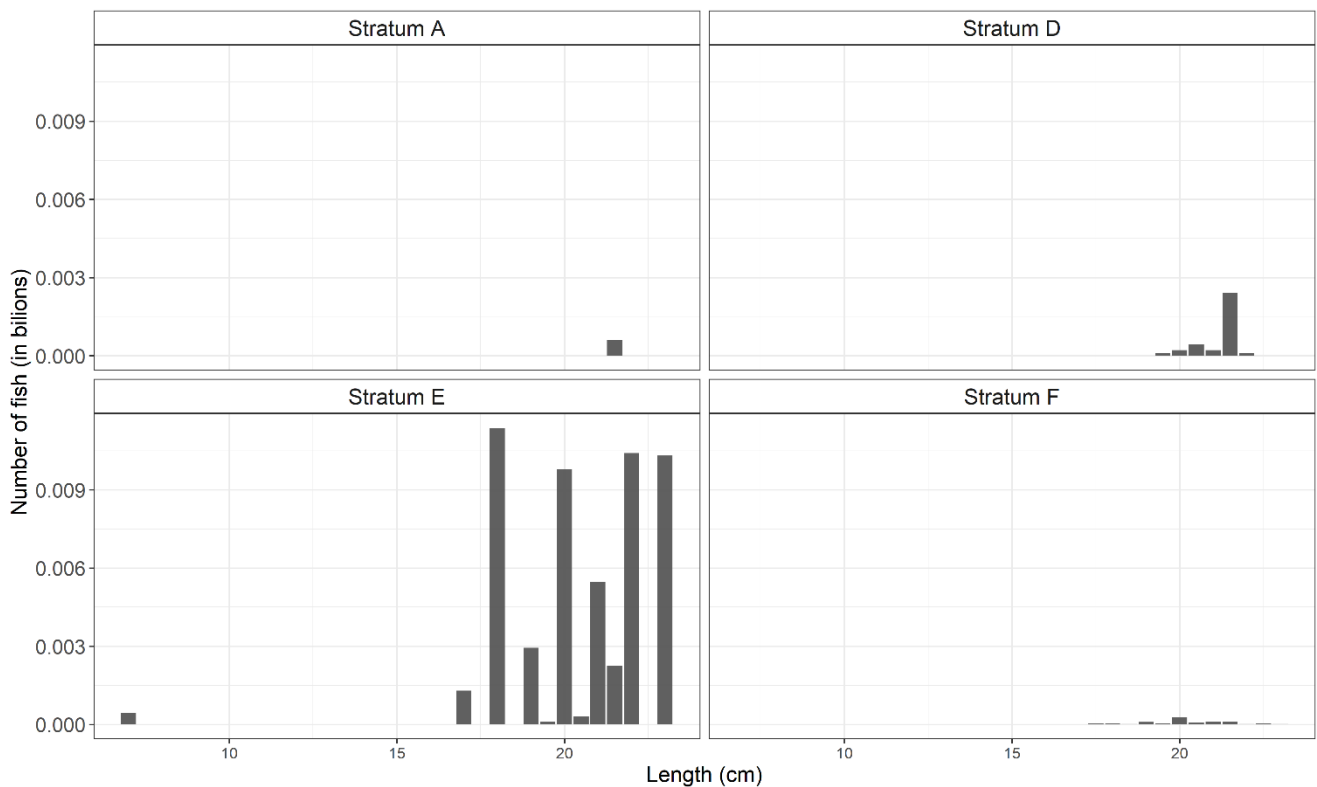


Figure B6: Length frequencies of chub mackerel per stratum. No chub mackerel were found in stratum B and C. Minor x-axis breaks represent 0.5 cm intervals.

Appendix C: Distribution maps

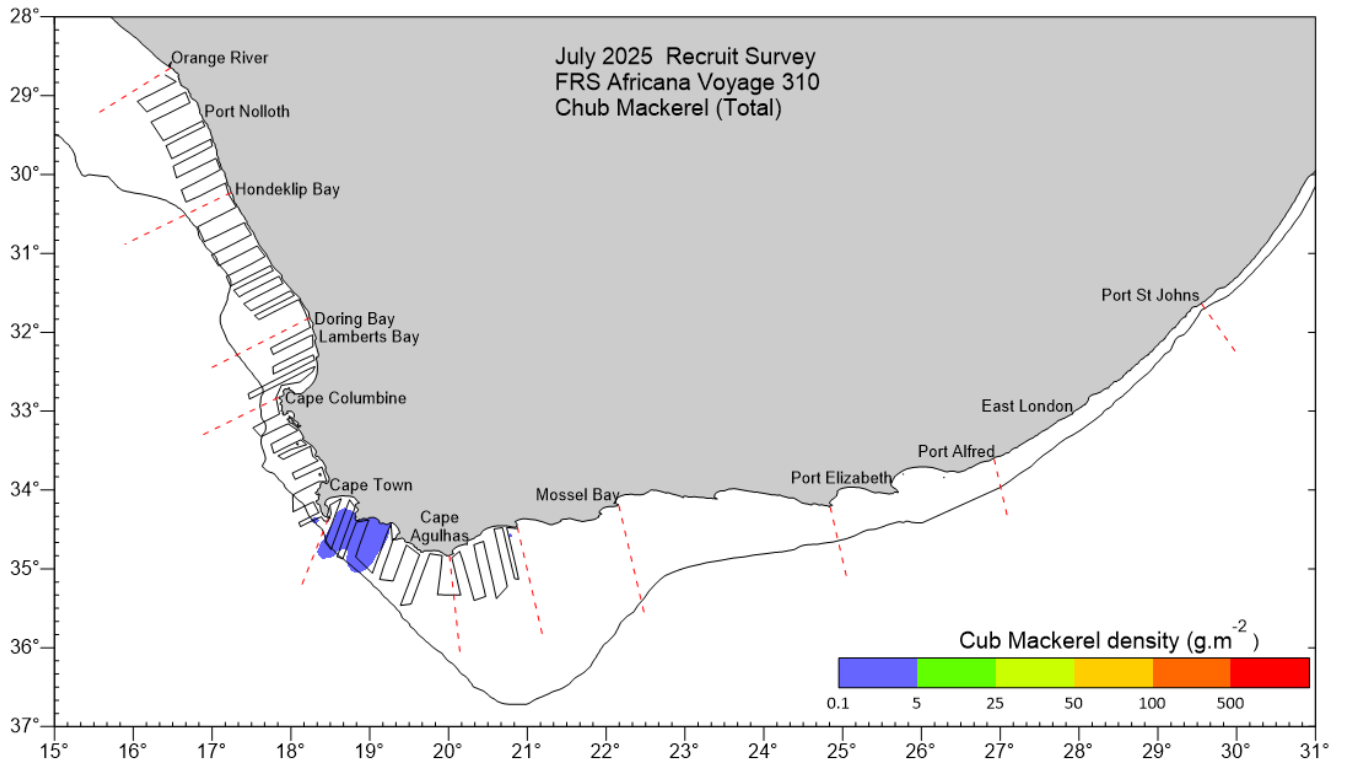


Figure C1: Distribution and relative abundance of all chub mackerel. Note the scale is 0.1 g.m^{-2} .

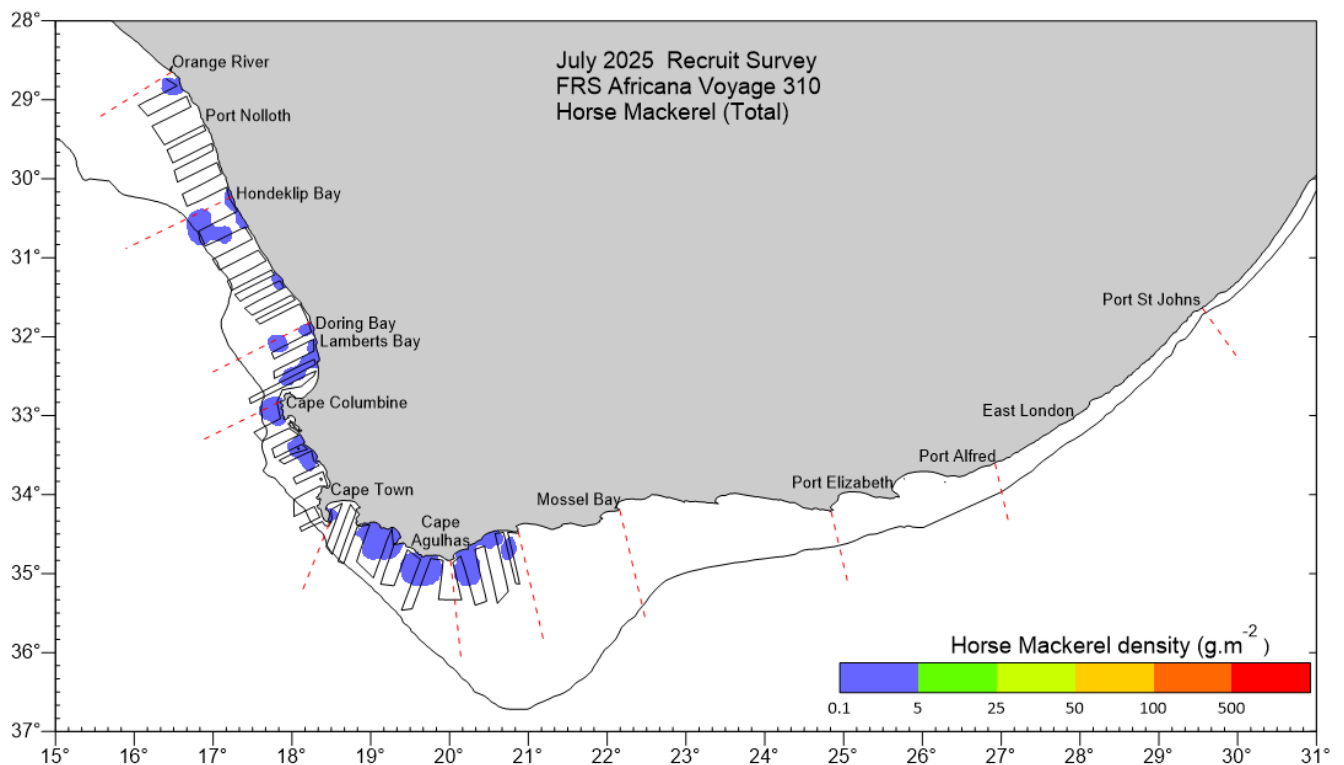


Figure C2: Distribution and relative abundance of all horse mackerel. Note that the density scale starts at 0.1 g.m^{-2} .

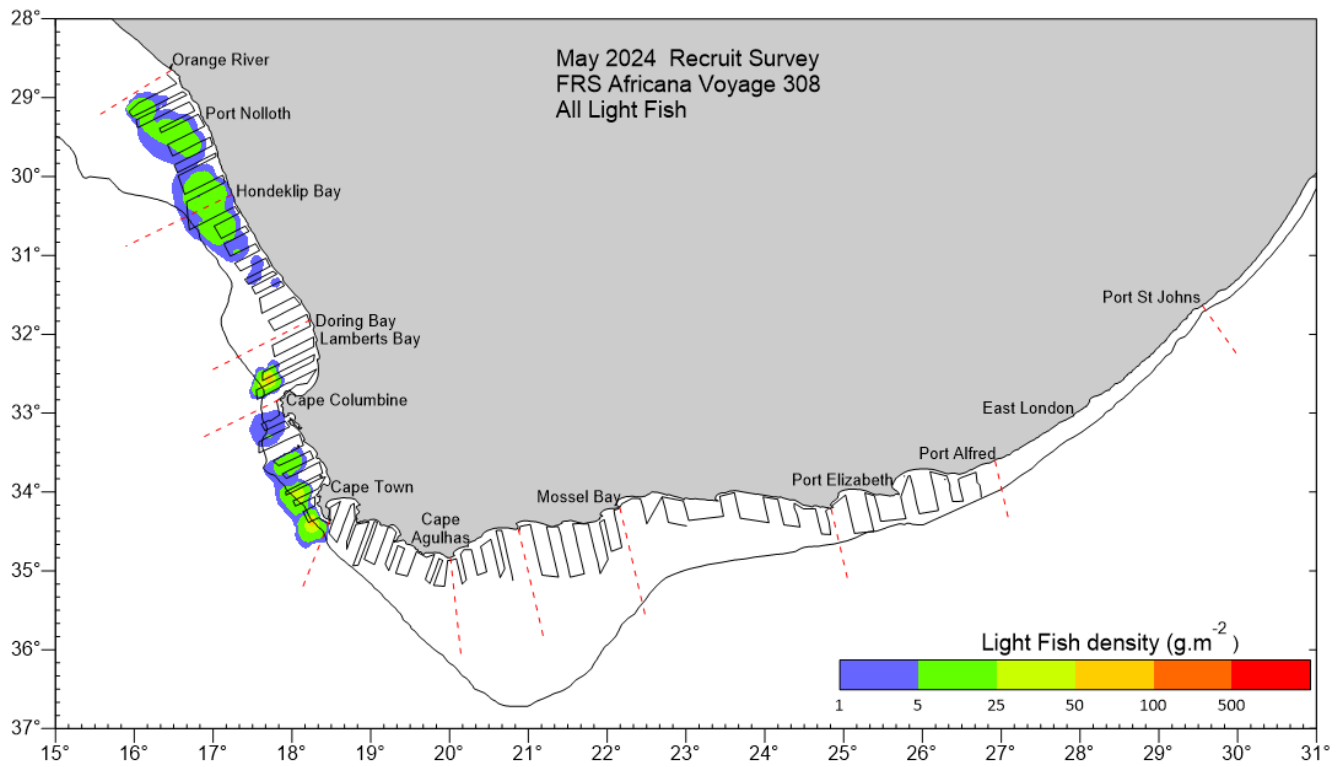


Figure C3: Distribution and relative abundance of all lightfish.

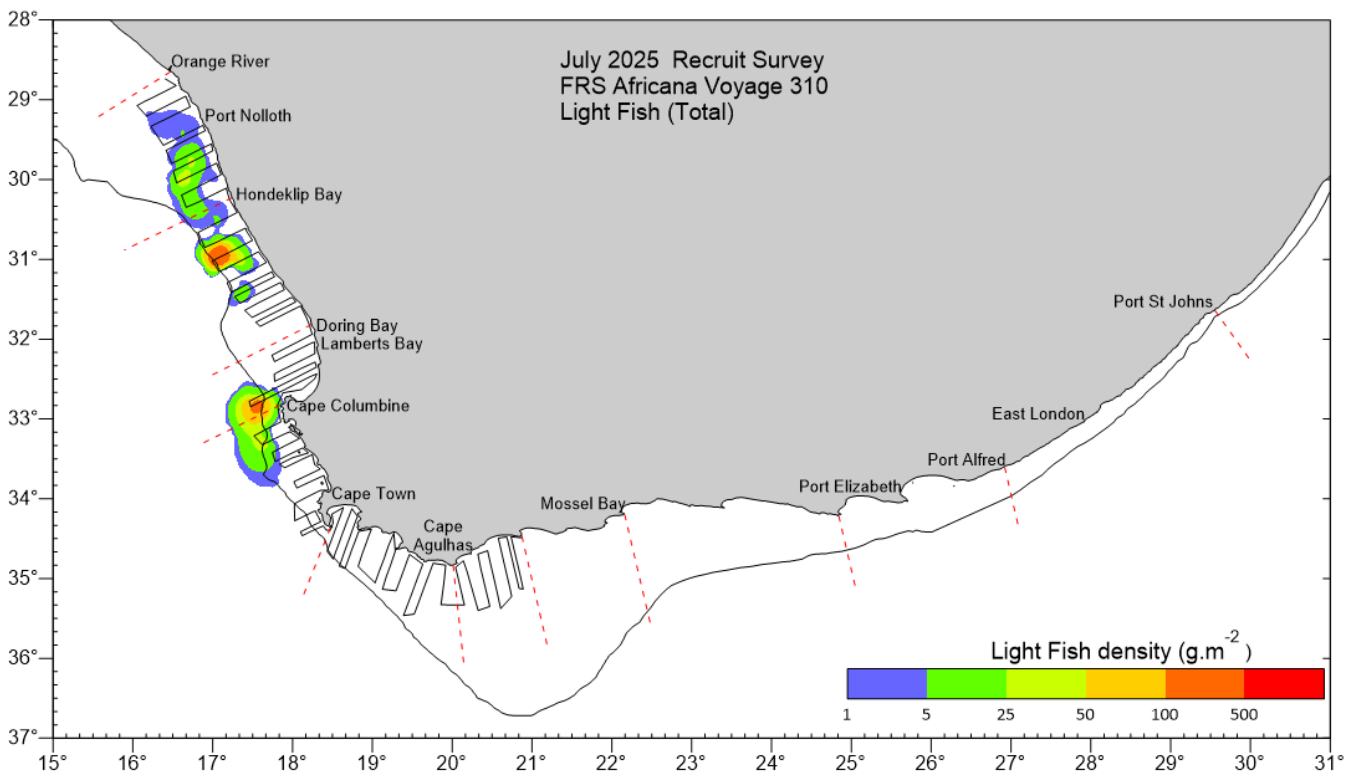


Figure C4: Distribution and relative abundance of all lanternfish.